



BOARD QUESTION PAPER : MARCH 2015

Notes:

- All questions are compulsory.
- Figures to the right indicate full marks.
- Answer to every question must be written on a new page.
- L.P.P. problem should be solved on graph paper.
- Log table will be provided on request.
- Write answers of Section – I and Section – II in one answer book.

Section – I

Q.1. Attempt any SIX of the following:**[12]**

- Express the following statement in symbolic form and write its truth value.
“If 4 is an odd number, then 6 is divisible by 3.” (2)
- Find the values of x and y , if
$$2 \begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$$
 (2)
- Find the value of ‘ k ’ if the function
$$f(x) = \frac{\tan 7x}{2x}, \quad \text{for } x \neq 0$$
$$= k, \quad \text{for } x = 0$$
is continuous at $x = 0$ (2)
- Find $\frac{dy}{dx}$ if $y = \cos^{-1}(\sqrt{x})$ (2)
- The price P for demand D is given as $P = 183 + 120D - 3D^2$.
Find D for which the price is increasing. (2)
- Evaluate: $\int \frac{1}{x(3 + \log x)} dx$ (2)
- If $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$ show that $A^2 - 3A + I = 0$ (2)
- Evaluate: $\int x \cos x dx$. (2)

Q.2. (A) Attempt any TWO of the following:**[6][14]**

- Prove that the following statement pattern is equivalent:
 $(p \vee q) \rightarrow r$ and $(p \rightarrow r) \wedge (q \rightarrow r)$ (3)
- Examine the continuity of the following function:
$$f(x) = \begin{cases} x^2 - x + 9, & \text{for } x \leq 3 \\ 4x + 3, & \text{for } x > 3 \end{cases}$$
 at $x = 3$ (3)
- Find $\frac{dy}{dx}$ if $y = \tan^{-1}\left(\frac{6x}{1 - 5x^2}\right)$ (3)

**(B) Attempt any TWO of the following:****[8]**

- i. Find the inverse of the following matrix by elementary row transformations if it exists.

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 0 & -2 & 1 \\ -1 & 3 & 0 \end{bmatrix} \quad (4)$$

- ii. Find area of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (4)

- iii. The expenditure E_c of a person with income I is given by $E_c = (0.000035)I^2 + (0.045)I$. Find marginal propensity to consume (MPC) and marginal propensity to save (MPS) when $I = 5000$. Also find A (average) PC and A (average) PS. (4)

Q.3. (A) Attempt any TWO of the following:**[6][14]**

- i. Express the truth of each of the following statements by Venn diagram:

- Some hardworking students are obedient.
- No circles are polygons.
- All teachers are scholars and scholars are teachers. (3)

- ii. If 'f' is continuous at $x = 0$, then find $f(0)$.

$$f(x) = \frac{15^x - 3^x - 5^x + 1}{x \tan x}, x \neq 0 \quad (3)$$

- iii. Find $\frac{dy}{dx}$ if $x = e^{2t}, y = e^{\sqrt{t}}$ (3)

(B) Attempt any TWO of the following:**[8]**

- i. Evaluate: $\int \frac{(1 + \log x)}{x(2 + \log x)(3 + \log x)} dx$ (4)

- ii. Evaluate: $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \cot x}$ (4)

- iii. A firm wants to maximize its profit. The total cost function is $C = 370Q + 550$ and revenue is $R = 730Q - 3Q^2$. Find the output for which profit is maximum and also find the profit amount at this output. (4)

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Section – I

Question 1 to 3 (based on section I) are given in our book STD XII (COMMERCE) MATHEMATICS AND STATISTICS - I

Section – II**Q.4. Attempt any SIX of the following:****[12]**

- The ratio of number of boys and girls in a school is 3 : 2. If 20 % of the boys and 30 % of the girls are scholarship holders, find the percentage of students who are not scholarship holders (2)
- Obtain crude death rates (C.D.R.) for city A and city B from the data given below:

Age group (in years)	City A		City B	
	Population	No. of deaths	Population	No. of deaths
Below 15	800	32	900	12
15 – 25	3000	12	1500	8
25 – 65	4800	48	4500	38
65 and above	1400	42	600	30

(2)

- Coefficient of rank correlation between x and y is 0.5 and $\sum d_i^2 = 42$. Assuming that no ranks are repeated, find the number of pairs of observations. (2)
- An agent charges 12 % commission on the sales. What does he earn if the total sale amounts to ₹ 36,000? What does the seller get? (2)
- Find the age standard death rate (S.D.R.) for the following data:

Age group (in years)	Population (in '000)	No. of deaths
0 – 10	11	240
10 – 20	12	150
20 – 60	9	125
60 and above	2	90

(2)

- Following table gives the age of husbands and age of wives.

Age of wives (in years)	Age of husbands (in years)			
	20–30	30–40	40–50	50–60
15 – 25	5	9	3	–
25 – 35	–	10	25	2
35 – 45	–	1	12	2
45 – 55	–	–	4	16
55 – 65	–	–	–	4



Find:

- a. The marginal frequency distribution of the age of husbands.
- b. The conditional distribution of the age of husbands when the age of wives lies between 25 – 35. (2)
- vii. The present worth of the sum of ₹ 5,830, due 9 months hence, is ₹ 5,500. Find the rate of interest. (2)
- viii. For a binomial distribution mean is 6 and variance is 2. Find n and p. (2)

Q.5. (A) Attempt any TWO of the following: (6) [14]

- i. For the following problem, find the sequence that minimizes total elapsed time (in hours) required to complete jobs on two machines M_1 and M_2 in the order $M_1 - M_2$. Also find the minimum elapsed time T.

Jobs	A	B	C	D	E
Machine M_1	5	1	9	3	10
Machine M_2	2	6	7	8	4

(3)

- ii. Mr. Natarajan and Mr. Gopalan are partners in the company having capitals in the ratio 4 : 5 and the profits received by them are in the ratio 5 : 4. If Mr. Gopalan invested capital in the company for 16 months, how long was Mr. Natarajan's investment in the company? (3)
- iii. From a lot of 25 bulbs of which 5 are defective a sample of 5 bulbs was drawn at random with replacement. Find the probability that the sample will contain
 - a. exactly 1 defective bulb
 - b. at least 1 defective bulb. (3)

(B) Attempt any TWO of the following: (8)

- i. Given the following table which relates to the number of parrots at age x , complete the life table for parrots.

x	0	1	2	3	4	5
l_x	1000	940	780	590	25	0

(4)

- ii. You are given the following information about advertising expenditure and sales:

	Advertisemet	
	Expenditure (₹ in lakh) (X)	Sales (₹ in lakh) (Y)
Arithmetic mean	10	90
Standard deviation	3	12

Correlation coefficient between X and Y = 0.8.

- a. Obtain the two regression equations.
- b. What is the likely sales when the advertising budget is ₹ 15 lakh?
- c. What should be the advertising budget if the company wants to attain sales target of ₹ 120 lakh? (4)
- iii. Electro Corp.Co. manufactures two electrical products: Air conditioners and Fans. The assembly process for each is similar in which both require a certain amount of wiring and drilling. Each air conditioner takes 4 hours for wiring and 2 hours for drilling. Each fan also takes 2 hours for wiring and 1 hour for drilling. During the next production period, 240 hours of wiring time are available and upto 100 hours of drilling time may be used. Each air-conditioner assembled may be sold for ₹ 2,000 profit and each fan assembled may be sold for ₹ 1,000 profit. Formulate this problem as an L.P.P. in order to maximize the profit.

**Q.6. (A) Attempt any TWO of the following:****(6)[14]**

- i. The equations given of the two regression lines are:

$$2x + 3y - 6 = 0 \text{ and } 5x + 7y - 12 = 0$$

Find:

- a. Correlation coefficient

b. $\frac{\sigma_x}{\sigma_y}$ (3)

- ii. Find graphical solution for the following system of linear inequations:

$$2x + 3y \geq 12, -x + y \leq 3, x \leq 4, y \geq 3$$
 (3)

- iii. The number of complaints which a bank manager receives per day is a Poisson random variable with parameter
- $m = 4$
- . Find the probability that the manager will receive

- a. only two complaints on any given day.

- b. at most two complaints on any given day

[Use $e^{-4} = 0.0183$] (3)

(B) Attempt any TWO of the following:**(8)**

- i. A warehouse valued at ₹ 10,000 contained goods worth ₹ 60,000. The warehouse was insured against fire for ₹ 4,000 and the goods to the extent of 90% of their value. A fire broke out and goods worth ₹ 20,000 were completely destroyed, while the remainder was damaged and reduced to 80% of its value. The damage to the warehouse was to the extent of ₹ 2,000. Find the total amount that can be claimed. (4)

- ii. In the following data, one of the values of Y is missing. Arithmetic means of X and Y series are 6 and 8 respectively.

X	6	2	10	4	8
Y	9	11	?	8	7

- a. Estimate the missing observation.

- b. Calculate correlation coefficient. (4)

- iii. A job production unit has four jobs A, B, C, D which can be manufactured on each of the four machines P, Q, R and S. The processing cost of each job is given in the following table :

Jobs	Machines			
	P	Q	R	S
	Processing Cost (₹)			
A	31	25	33	29
B	25	24	23	21
C	19	21	23	24
D	38	36	34	40

How should the jobs be assigned to the four machines so that the total processing cost is minimum? (4)